

Figure 5.7.1(a) Suction Stroke

ii) Compression Stroke

In this stroke, the piston moves up from bottom dead centre to top dead centre. During this stroke, both inlet and exhaust valves are closed. The air drawn into the cylinder during suction stroke is entrapped inside the cylinder and compressed due to upward movement of the piston. The fuel is injected at the end of the compression stroke by the injector and the fuel ignites.



Figure 5.7.1(b) Compression Stroke

iii) Power Stroke

The hot gases which are produced due to ignition of fuel during compression stroke and compressed air now expand adiabatically, in the cylinder pushing the piston down towards the BDC. This downward movement of the piston is converted into rotary motion of the crankshaft and hence work is done. During expansion, the pressure and the temperature reduce. In this stroke, both inlet and exhaust valve remain closed.



Figure 5.7.1(c) Power Stroke

iv) Exhaust Stroke

At the end of the power stroke, exhaust stroke will start. In this stroke, the piston again moves upward. The exhaust valve opens, while inlet valve is closed. A greater part of the burnt fuel gases escapes due to their own expansion. The upward movement of the piston pushes the remaining gases out through the open exhaust valve. At the end of an exhaust stroke, the exhaust valve closes and the cycle is thus completed and suction stroke of next cycle is started.

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